

Fast and Stable Elastic Dynamics with Large-scale Contact Simulation

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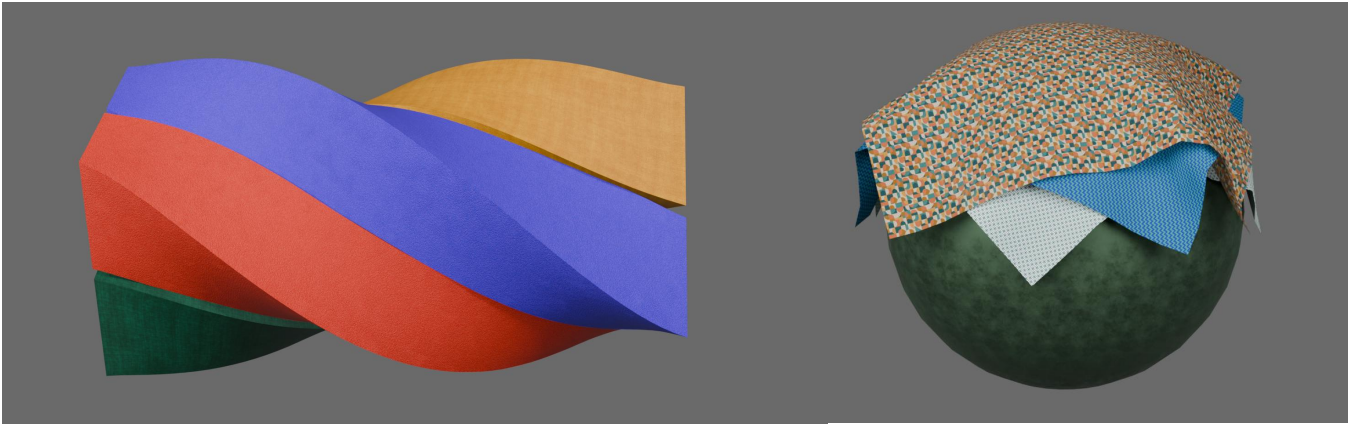


Figure 1: Examples of contact-rich scenarios with large contact areas. Left: four volumetric bars twisted together. Right: three stacked cloth layers, where the number of contact constraints even exceeds the mechanical DoFs. Our method efficiently handles both cases, achieving real-time performance with very low GPU memory consumption.

CCS Concepts

• **Computing methodologies** → **Physical simulation; Real-time simulation.**

Keywords

Physics-based animations, Real-time simulations, GPU parallelization

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1 Introduction

Physics-based simulation plays a central role in computer graphics and interactive applications, where accurate modeling of deformable materials is essential for realism and reliability. Among the many challenges in this domain, contact-rich scenarios—such as cloth–cloth, cloth–body, or granular interactions—remain particularly difficult. These settings demand not only robustness against complex collision configurations but also computational efficiency to support real-time or large-scale simulations.

Classical approaches for contact response, such as penalty methods, offer simplicity and efficiency but suffer from soft constraint enforcement and parameter sensitivity. In contrast, Lagrange multiplier formulations—including non-smooth Newton methods [Macklin et al. 2019] with NCP function constraints—enforce contact conditions strictly and often achieve faster convergence. Building on this idea, [Zeng et al. 2025] proposed a new algorithm that integrates NCP constraints into local-global solvers [Bouaziz et al. 2014], thereby addressing both elastic dynamics and contact response within a fully GPU-parallelized implicit scheme. However, the fully implicit treatment of contact inevitably produces dense contact system operators, making memory consumption and computation prohibitively expensive in contact-rich simulations.

Table 1: Simulation statistics comparing the baseline [Zeng et al. 2025] and our method. Improvements are highlighted in GPU memory consumption as well as performance in the Schur-complement (Schur.) and constraint solver (Cst. Solve).

Example	Mesh		Elasticity		Contact	Method	Contact Matrices Memory	Time cost (ms)		
	Vert.	Elem.	Constitutive	$\rho(kg/m^3), E(Pa), \nu$				Schur.	Cst. Solve	Total
Fig.1 left	5k×4	24k×4	Neo-Hookean	1000, 1E+5, 0.45	9k	[Zeng et al. 2025]	1.1 GB	45.21	32.40	267.46
						Ours	8.0MB	2.00	1.15	56.12
Fig.1 right	2k×3	4k×3	Neo-Hookean	1000, 1E+5, 0.45	13k	[Zeng et al. 2025]	2.6 GB	144.68	23.32	292.88
						Ours	8.4 MB	2.08	0.72	22.82

To overcome this limitation, we propose a lightweight approximation scheme that significantly reduces the density of the contact system. Our method draws inspiration from splitting strategies [Ly et al. 2020] and adapts them to the NCP constrained dynamics, enabling efficient simulations in contact-rich scenarios.

2 Method

We recall the assembled system in [Zeng et al. 2025]:

$$\begin{bmatrix} \mathbf{A} & -\mathbf{J}^T \\ \mathbf{J} & \mathbf{E} \end{bmatrix} \begin{bmatrix} \mathbf{q}^{k+1} \\ h^2 \Delta \lambda \end{bmatrix} = \begin{bmatrix} \mathbf{g} \\ \mathbf{h} \end{bmatrix} \quad (1)$$

A standard approach to solving this system is through the Schur-complement $\mathbf{J}\mathbf{A}^{-1}\mathbf{J}^T + \mathbf{E}$. In [Zeng et al. 2025], the sparse inverse $\mathbf{S} = \mathbf{L}^{-1}$ is provided, which allows the Schur complement to be efficiently evaluated using matrix multiplications. However, as revealed in [Zeng et al. 2025], the system inverse $\mathbf{A}^{-1}$ is generally fully dense. Consequently, the Delassus operator $\mathbf{W} = \mathbf{H}\mathbf{A}^{-1}\mathbf{H}^T$ is definitely fully dense in this case. While this is manageable in low-contact scenarios, where computation and memory costs remain modest, it becomes prohibitive in contact-rich settings due to performance degradation and GPU memory limitations.

To address this issue, we draw inspiration from the splitting scheme in [Ly et al. 2020]. Specifically, we approximate

$$\mathbf{A}\mathbf{q}^{k+1} \approx \mathbf{M}\mathbf{q}^{k+1} + \mathbf{D}\mathbf{q}^k \quad (2)$$

where $\mathbf{M}$ denotes the mass matrix, and $\mathbf{D} = h^2 \sum_i w_i \mathbf{G}_i^T \mathbf{G}_i$ assembles the pre-defined constraints, such that $\mathbf{A} = \mathbf{M} + \mathbf{D}$. This leads to the following approximation of Equation (1):

$$\begin{bmatrix} \mathbf{M} & -\mathbf{J}^T \\ \mathbf{J} & \mathbf{E} \end{bmatrix} \begin{bmatrix} \mathbf{q}^{k+1} \\ h^2 \Delta \lambda \end{bmatrix} = \begin{bmatrix} \mathbf{g} - \mathbf{D}\mathbf{q}^k \\ \mathbf{h} \end{bmatrix} \quad (3)$$

The resulting formulation modifies the pipeline in [Zeng et al. 2025] and is summarized in Algorithm 1, with the key differences highlighted.

Through this reformulation, we obtain a lightweight variant of the method in [Zeng et al. 2025]. The sparsity of $\mathbf{W}$ is significantly improved, though at the expense of reduced coupling capacity, since the fully implicit contact scheme is relaxed to a semi-implicit one. Nevertheless, this reduction in density enables contact-rich simulations, such as multi-layer cloth dynamics, at real-time performance levels.

3 Results and Conclusion

Our GPU-based method outperforms [Zeng et al. 2025], reducing memory from gigabytes to megabytes and achieving 5–10×

ALGORITHM 1: Lite local-global solver with NCP constraints, adapted from [Zeng et al. 2025] with a mass-constraint splitting to reduce the density of the Delassus operator.

```

L = Cholesky(A);
S = L-1I;
while simulation do
    Perform collision detection;
    W = HM-1HT;
    q̃ = qt + hvt + h2M-1fext;
    for k ∈ {0, ..., n} do
        pik = project(Giqk); // local step
        b = Mq̃ + h2 ∑ wi GiT pik;
        Evaluate Ω, J, E, g, h;
        Δλ =  $\frac{1}{h^2} \left( \mathbf{\Omega} \mathbf{W} \mathbf{\Omega}^T + \mathbf{E} \right)^{-1} \left( \mathbf{h} - \mathbf{J} \mathbf{M}^{-1} (\mathbf{g} - \mathbf{D} \mathbf{q}^k) \right)$ ;
        λk+1 = λk + Δλ;
        qk+1 = STS(b + h2JTλk+1);
    end
    qt+h = qk+1;    vt+h =  $\frac{1}{h}(\mathbf{q}^{k+1} - \mathbf{q}_t)$ ;
end

```

speedups for real-time contact-rich simulations. Unlike [Ly et al. 2020], it handles general vertex-face and edge-edge contacts via an NCP-based conjugate residual solver for efficient parallel convergence. Our solver enables large-scale, real-time simulations and parallel robotic training with multi-layer cloth and deformable objects, supporting interactive design and high-throughput data generation for manipulation tasks.

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